

Curriculum Vitae

Ke Zhang

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EDUCATION

University of California, Riverside

Ph.D. in Mathematics, GPA: 3.66/4.0

Expected May 2028

Northeastern University, College of Science

MSc in Mathematics, GPA: 3.8/4.0

Courses: Algebra 1&2, Computer Algebra, Machine Learning

Boston, Massachusetts

May 2023

Arizona State University, School of Mathematical and Statistical Sciences

MA in Mathematics (transferred)

Courses: General topology, algebraic topology

Tempe, Arizona

08/2021-12/2021

Southwest Minzu University, School of Economics

Bachelor of Economics, GPA: 3.44/4.0

Major: Financial Engineering

Chengdu, Sichuan, China

06/2021

RESEARCH EXPERIENCE

Understanding Tool-Augmented Agents for Lean Formalization: A Factorial Analysis

ICML 2026 — Under Review (first author); extended from NeurIPS 2025 Workshop on AI+Math — Accepted

- Proposed an agentic framework for Lean 4 formalization that replaces one-shot generation with iterative refinement via three tool categories — symbol retrieval, expert drafting (fine-tuned Herald), and Lean REPL feedback — lifting compilation success from 26.0% (GPT-5.2 one-shot) to 89.5% and faithfulness from 28.0% to 60.5%.
- Curated a new benchmark of 400 graduate-level theorems spanning Real Analysis, Complex Analysis, Topology, and Algebra, and designed a full factorial experiment to quantify the marginal and interaction effects of each tool category on compilation and semantic faithfulness.
- Built the full system end-to-end: Dockerized Lean 4 + Mathlib sandbox, LLM-as-Judge faithfulness evaluator (GPT-5.2 Medium), and an automated pipeline executing 400+ theorems under each tool configuration while logging REPL traces, tool calls, and compile outcomes for cross-model comparison.

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Learning Parameterized Nonlinear Elasticity on Curved Surfaces

ICLR 2026 — Accepted (co-author; led training)

- Designed and executed the training pipeline for a parameter-conditioned physics-informed neural network (PINN) that learns families of nonlinear elastic equilibria on curved manifolds, achieving $\sim 10^{-3}$ loss across a 3D parameter space (β , q , vp).
- Implemented a two-stage learning rate schedule (200k steps at 10^{-4} , 100k at 10^{-5}) in PyTorch on dual NVIDIA RTX A6000 GPUs, with the trained model generalizing to held-out parameter combinations not seen during training.
- Conducted systematic ablation studies on collocation density (1k–7k points) and architecture (2–6 layers, 160–240 neurons), identifying failure modes and stable configurations.

Agent-Based Code Repair System for Gurobi Optimization Models

AAAI 2025 Workshop — Accepted

- Developed a benchmark suite of 26 Gurobi optimization problems (LP, ILP, MIP, QCP, and combinatorial/logistics applications) and implemented 260 unit tests covering feasibility, infeasibility,

optimality correctness, and parameter validation.

- Built an automated LLM-based bug generation system, creating reproducible perturbations in Gurobipy code to form controlled debugging cases.
- Constructed a RAG system over the full Gurobi Python API reference, enabling grounded retrieval of function signatures, parameters, model attributes, to reduce hallucination and improve repair accuracy.
- Performed fractional factorial analysis to study the effects of different tool combinations on repair success.

PROFESSIONAL ACTIVITIES

MSRI summer school

Moraga, California

Topological Methods for the Discrete Mathematician

07/2022 – 08/2022

- Learned how to use equivariant topology, especially the Borsuk–Ulam theorem, to solve problems in combinatorics and geometry
- Discussed open problems in related fields, such as the mass partition problem, with other participants

TEACHING ACTIVITIES

Teaching Assistant, College of Science, University of California Riverside

MATH 005A

Fall 2025

MATH 009B

Spring 2025

MATH 120

Summer 2024

MATH 009A

Summer 2024

MATH 006A

Winter 2024

MATH 007A

Fall 2024

Teaching Assistant, College of Science, Northeastern University

Group Theory (MATH 3175)

Fall 2022

Group Theory (MATH 3175)

Spring 2023

COMPUTER SKILLS

Python, C, C++, Docker